Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

For a real gas $(\text{mol.mass} = 60)$ if density at critical point is $0.80\ \text{g/cm}^3$ and its $T_c = \frac{4\times10^5}{821}\ \text{K}$, then van der waals' constant a (in atm $\text{L}^2\ \text{mol}^{-2}$) is

(A) 0.3375

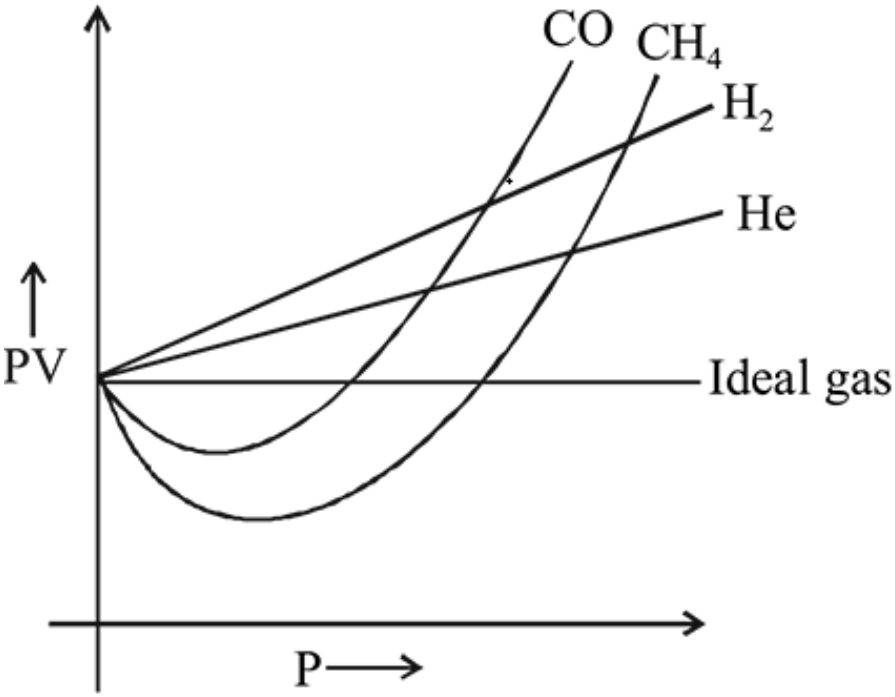
(B) 3.375

(C) 1.68

(D) 0.025

Q.2

In case of CO and CH_4 curve goes to minima then increases with increase in pressure but in case of H_2 and He the curve is linear because:



(A) Intermolecular interactions for H_2 and He are very low.

(B) Molecular size or atomic size for H_2 and He is small.

(C) Both (a) and (b)

(D) Neither (a) nor (b)

Q.3

van der Waal's equation for a gas is stated as,

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$\mathrm{P}=\frac{\mathrm{nRT}}{\mathrm{V}-\mathrm{nb}}-\mathrm{a}\left(\frac{\mathrm{n}}{\mathrm{V}}\right)^2$

This equation reduces to the perfect gas equation, $\mathrm{P}=\frac{\mathrm{nRT}}{\mathrm{V}}$ when,

(A) temperature is sufficient high and pressure is low.

(B) temperature is sufficient low and pressure is high.

(C) both temperature and pressure are very high.

(D) both temperature and pressure are very low.

Q.4

For real gases van der Waals equation is written as

$\left(\mathrm{p}+\frac{\mathrm{an}^2}{\mathrm{V}^2}\right)(\mathrm{V}-\mathrm{nb})=\mathrm{nRT}$

where ' a ' and ' b ' are van der Waals constants. Two sets of gases are :

(I) $\mathrm{O}_2, \mathrm{CO}_2, \mathrm{H}_2$ and He

(II) $\mathrm{CH}_4, \mathrm{O}_2$ and H_2

The gases given in set-I in increasing order of ' b ' and gases given in set-II in decreasing order of ' a ', are arranged below. Select the correct order from the following:

(A) (I) $\mathrm{He}<\mathrm{H}_2<\mathrm{CO}_2<\mathrm{O}_2$
(II) $\mathrm{CH}_4>\mathrm{H}_2>\mathrm{O}_2$

(B) (I) $\mathrm{O}_2<\mathrm{He}<\mathrm{H}_2<\mathrm{CO}_2$
(II) $\mathrm{H}_2>\mathrm{O}_2>\mathrm{CH}_4$
(II) $\mathrm{CH}_4>\mathrm{O}_2>\mathrm{H}_2$

(C) (I) $\mathrm{He}<\mathrm{H}_2<\mathrm{O}_2<\mathrm{CO}_2$

(D) (I) $\mathrm{H}_2<\mathrm{O}_2<\mathrm{He}<\mathrm{CO}_2$ (II) $\mathrm{O}_2>\mathrm{CH}_4>\mathrm{H}_2$

Q.5

1 mole of SO_2 occupies a volume of $350 \mathrm{ml}$ at $300 \mathrm{~K}$ and 50 atm pressure. What is the compressibility factor of the gas.

(A) 1.0

(B) 0.09

(C) 2.12

(D) 0.71

Q.6

The rate of diffusion of SO_2 and O_2 are in the ratio

(A) $1: \sqrt{2}$

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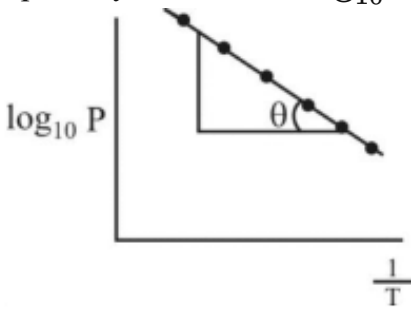
(B) $1:32$

(C) $1:2$

(D) $1:4$

Q.7

Graphically variation of $\log_{10} P$ with $\frac{1}{T}$ is shown.



If $\tan\theta$ (slope of the line) of the given liquid is $= -2\ \mathrm{K}$, latent heat of vaporisation of the given liquid is:

(A) $2.303\mathrm{calmol}^{-1}$

(B) $4.606\mathrm{calmol}^{-1}$

(C) $9.212\mathrm{calmol}^{-1}$

(D) $4.00\mathrm{calmol}^{-1}$

Q.8

If Z is a compressibility factor, van der Waals equation at
low pressure can be written as:

(A) $Z = 1 + \frac{RT}{Pb}$

(B) $Z = 1 - \frac{a}{VRT}$

(C) $Z = 1 - \frac{Pb}{RT}$

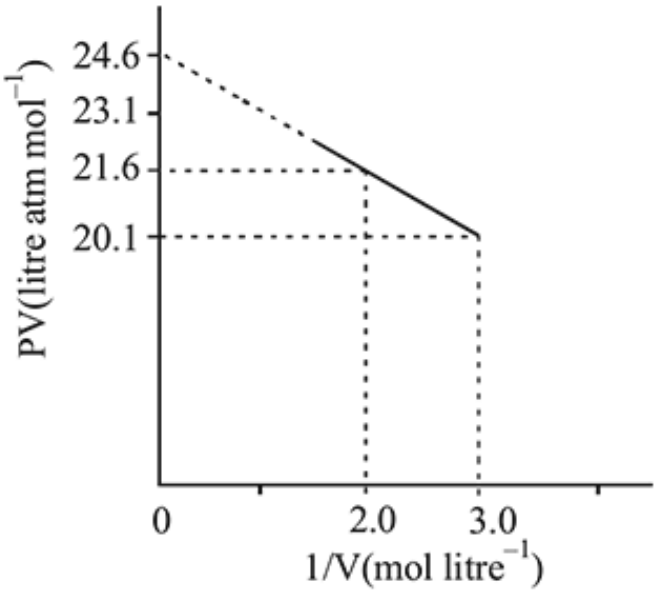
(D) $Z = 1 + \frac{Pb}{RT}$

Q.9

For one mole of a van der Waals gas when $b = 0$ and $\mathrm{T} = 300\ \mathrm{K}$, the PV vs, $1/\mathrm{V}$ plot is shown below. The

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value of the van der Waals constant a (atm. $\text{lit}^2\ \text{mol}^{-2}$) is :



(A) 1.5

(B) 2.5

(C) 3.5

(D) 1.9

Q.10

A manifestation of surface tension is :

(A) rise of liquid in a capillary tube

(B) spherical shape of liquid drops

(C) upward movement of water in soils

(D) All the above

Q.11

If 10^{-4}dm^3 of water is introduced into a 1.0dm^3 flask at $300\ \text{K}$, how many moles of water are in the vapour phase when equilibrium is established? (Given : Vapour pressure of H_2O at $300\ \text{K}$ is $3170\ \text{Pa}; \text{R} = 8.314\ \text{J K}^{-1}\ \text{mol}^{-1}$)

(A) 5.56×10^{-3} mol

(B) 1.53×10^{-2} mol

(C) 4.46×10^{-2} mol

(D) 1.27×10^{-3} mol

Q.12

Following table represents critical temperature of some gases. Arrange these gases in their increasing order of

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liquifaction. \begin{array}{|c|c|c|c|}
\hline Gas & \mathrm{H}_{2} & \mathrm{He} & \mathrm{N}_{2} & \mathrm{O}_{2} \\
\hline \mathrm{T}_{\mathrm{c}} / \mathrm{K} & 33.2 & 5.3 & 126 & 154.3 \\
\hline
\end{array}

(A) $\mathrm{He} < \mathrm{N}_2 < \mathrm{H}_2 < \mathrm{O}_2$

(B) $\mathrm{H}_2 < \mathrm{He} < \mathrm{N}_2 < \mathrm{O}_2$

(C) $\mathrm{He} < \mathrm{H}_2 < \mathrm{N}_2 < \mathrm{O}_2$

(D) $\mathrm{O}_2 < \mathrm{N}_2 < \mathrm{H}_2 < \mathrm{He}$

Q.13

The inversion temperature $T_i(\mathrm{K})$ of hydrogen is (given van der Waal's constants a and b are $0.244\ \mathrm{atm\ L^2\ mol^{-2}}$ and $0.027\ \mathrm{L\ mol^{-1}}$ respectively)

(A) 440

(B) 220

(C) 110

(D) 330

Q.14

Which of the following expressions for van der Waals constant a is not correct?

(A) $a = 27 p_{\mathrm{c}} b^2$

(B) $a = 3 p_{\mathrm{c}} V_{\mathrm{c}}^2$

(C) $a = 27\left(RT_{\mathrm{c}}\right)^2 / 64 p_{\mathrm{c}}$

(D) $a = 27 R V_{\mathrm{c}} T_{\mathrm{c}} / 8$

Q.15

Which of the following expressions between critical constants and van der Waals constants is not correct?

(A) $T_{\mathrm{c}} = 8a/(27Rb)$

(B) $p_{\mathrm{c}} = a\left(3b^2\right)$

(C) $p_{\mathrm{c}} V_{\mathrm{c}} = (3/8) RT_{\mathrm{c}}$

(D) $V_{\mathrm{c}} = 3b$

Answers & Solutions

Q.1 Answer:

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3.375

Solution:

$\mathrm{V_c} = \frac{60}{0.80} = 75\ \mathrm{cm^3\ mol^{-1}}$
$\mathrm{b} = \frac{\mathrm{V_c}}{3} = 25\ \mathrm{cm^3\ mol^{-1}}; = 0.25\ \mathrm{L\ mol^{-1}}$
$\therefore \mathrm{T_c} = \frac{8\mathrm{a}}{27\mathrm{Rb}}$
$\frac{4\times10^5}{821} = \frac{2.45}{0.425} = 5.76\mathrm{gL^{-1}}$
$\Rightarrow \mathrm{a} = 3.375$

Q.2 Answer:

Both (a) and (b)

Solution:

Due to small size of these species $(\mathrm{H_2}$ and He) intermolecular interactions (van der Waal forces) are very low, therefore it is difficult to compress these.

Q.3 Answer:

temperature is sufficient high and pressure is low.

Solution:

Given $P = \frac{nRT}{\mathrm{v}-nb} - a\left(\frac{n}{\mathrm{v}}\right)^2$
Which can also be written as
$\left[P + \frac{n^2a}{V^2}\right](V - nb) = nRT$
At low pressure and high temperature the effect of $\frac{a}{V^2}$ **and** b **is negligible hence** $PV = nRT$

Q.4 Answer:

(I) $\mathrm{He} < \mathrm{H_2} < \mathrm{O_2} < \mathrm{CO_2}$

Solution:

Values of van der Waals Constants for Some Common Gases
\begin{array} \ \text{Gas} & a \left(\frac{\mathrm{atm}\ \mathrm{L}^2}{\mathrm{~mol}^2}\right) & b \left(\frac{\mathrm{L}}{\mathrm{mol}}\right)\\ \ \hline \mathrm{He} & 0.034 & 0.0237\\ \ \mathrm{H}_2 & 0.244 & 0.0266\\ \ \mathrm{N}_2 & 1.39 & 0.0391\\ \ \mathrm{O}_2 & 1.36 & 0.0318\\ \ \mathrm{Cl}_2 & 6.49 & 0.0562\\ \ \mathrm{CO}_2 & 3.59 & 0.0427\\ \ \mathrm{CH}_4 & 2.25 & 0.0428\\ \ \end{array}

Q.5 Answer:

0.71

Solution:

$\mathrm{P} = 50\ \mathrm{atm} \quad \mathrm{V} = 350\mathrm{ml} = 0.350$ **litre**
$\mathbf{n = 1}$ **mole**

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$\mathbf{T} = 300\ \mathrm{K} \because \mathrm{Z} = \frac{\mathrm{PV}}{\mathrm{nRT}}$
$\therefore Z = \frac{50 \times 0.350}{1 \times 0.082 \times 300} = 0.711$
Thus SO_2 is more compressible than expected from ideal behavior

Q.6 Answer:

$1 : \sqrt{2}$

Solution:

$$\frac{\mathrm{r_{SO_2}}}{\mathrm{r_{O_2}}} = \sqrt{\frac{\mathrm{M_{O_2}}}{\mathrm{M_{sO_2}}}} = \sqrt{\frac{32}{64}} = \frac{1}{\sqrt{2}}$$

Q.7 Answer:

$9.212 \mathrm{calmol}^{-1}$

Solution:

By Clausius-Clapeyron equation.
$\log_{10} \mathbf{P} = \log_{10}\ \mathrm{A} - \frac{\Delta \mathrm{H}}{2.303 \mathrm{RT}}$
Slope $\left(-\frac{\Delta H}{2.303 R}\right) = -2$
$\Delta \mathrm{H} = 2.303 \mathrm{R} \times 2 \ = 4.606 \times r = 4.606 \times 2 = 9.212$

Q.8 Answer:

$Z = 1 - \frac{a}{VRT}$

Solution:

Compressibility factor $(Z) = \frac{PV}{RT}$ **(For one mole of real gas) van der Waals equation**

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

At low pressure, volume is very large and hence correction term b can be neglected in comparison to very large volume of V.
i.e. $V - b \approx V$

$$\left(P + \tfrac{a}{V^2}\right) V = RT$$
$$PV + \tfrac{a}{V} = RT$$
$$PV = RT - \tfrac{a}{V}$$
$$\tfrac{PV}{RT} = 1 - \tfrac{a}{VRT}$$

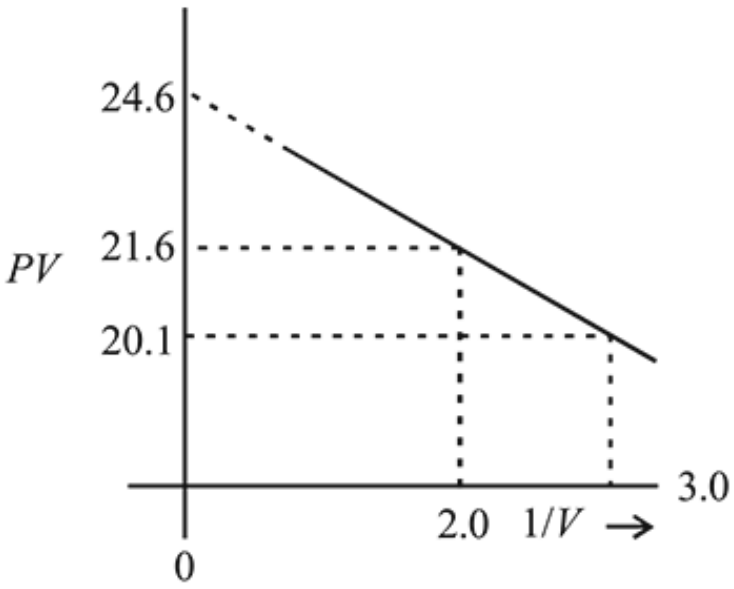
Hence, $Z = 1 - \frac{a}{VRT}$

Q.9 Answer:

1.5

Solution:

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$\left(P+\frac{a}{V^2}\right)(V)=RT$
$PV+a/V=RT$
$PV=RT-\frac{a}{V}$
$y=RT-a(x)$
So, slope $=-a=\frac{21.6-20.1}{2-3}=-a=-1.5$
$a=1.5$

Q.10 Answer:

All the above

Solution:

All the given phenomenon occurs due to surface tension

Q.11 Answer:

1.27×10^{-3} mol

Solution:

From the ideal gas equation :

$\mathrm{PV=nRT}$

or $\mathrm{n}=\frac{\mathrm{PV}}{\mathrm{RT}}=\frac{3170\times10^{-3}}{8.314\times300}=1.27\times10^{-3}$

Q.12 Answer:

$\mathrm{He}<\mathrm{H_2}<\mathrm{N_2}<\mathrm{O_2}$

Solution:

More will be critical temperature easier is the liquifaction of the gas. Hence correct order will be

$\mathrm{He}<\mathrm{H_2}<\mathrm{N_2}<\mathrm{O_2}$

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Q.13 Answer:

220

Solution:

Gases become cooler during Joule Thomson's expansion onlyif theyare belowa certain temperature known as inversion temperature (T_i). The inversion temperature is characteristic of each gas and is given by

$T_i = \frac{2a}{bR}$**, where R is gas constant Given**

$$a = 0.244 \text{ atm L}^2 \text{ mol}^{-2}$$
$$b = 0.027 \text{ L mol}^{-1}$$
$$R = 0.0821 \text{ L atmdeg}^{-1} \text{ mol}^{-1}$$
$$\therefore T_i = \frac{2 \times 0.244}{0.027 \times 0.0821} = 220 \text{ K}$$

Q.14 Answer:

$a = 27RV_cT_c/8$

Solution:

We have

$$a = 3p_cV_c^2 = 3p_c(3b)^2 = 27p_cb^2; \quad a = 3p_cV_c^2 = 3p_c\left(\frac{3RT_c}{8p_c}\right)^2 = \frac{27(RT_c)^2}{64p_c}$$
$$a = 3p_cV_c^2 = 3\left(\frac{3RT_c}{8V_c}\right)\left(V_c^2\right) = \frac{9}{8}RT_cV_c$$

Q.15 Answer:

$p_c = a\left(3b^2\right)$

Solution:

$p_c = a\left(3b^2\right)$

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